

What's in a Name

Fuzzy matching techniques
for proper nouns

aka.ms/foundry-discord-rn



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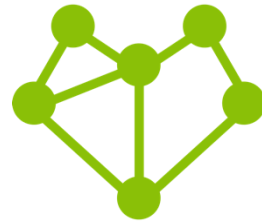
🌸 [@reeneenoble.bsky.social](https://reeneenoble.bsky.social)

Hi, I'm Renee!



Microsoft

Senior Python Cloud Advocate



Girls' Programming Network
CEO



Sydney Python Meetup
Co-organiser

Agenda

01 **My data dilemma**

02 **Fake your way through Sydney school names, with Gen AI**

03 **Comparing fuzzy matching libraries, for humans**

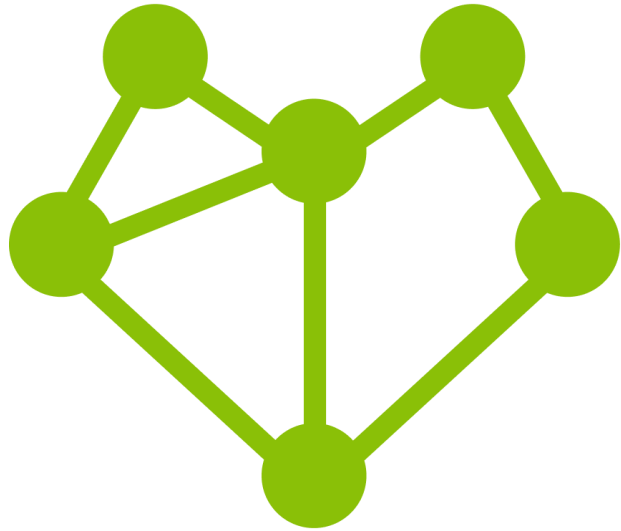
04 **My hybrid approach to fuzzy matching**

05 **What will I match next?**

My Data

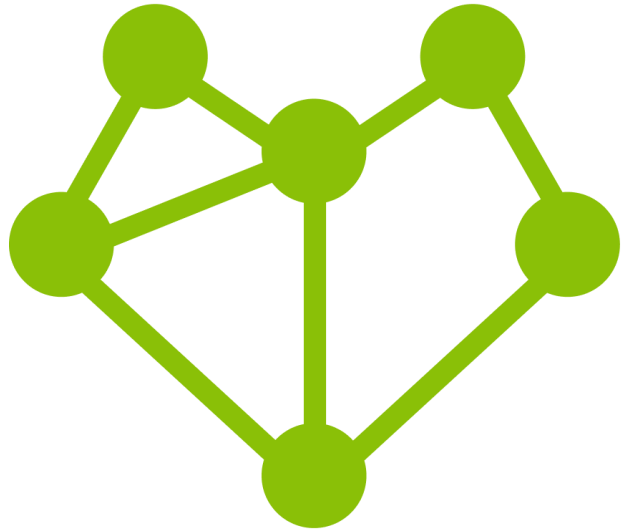
Dilemma

Let's talk about...



Girls' Programming Network

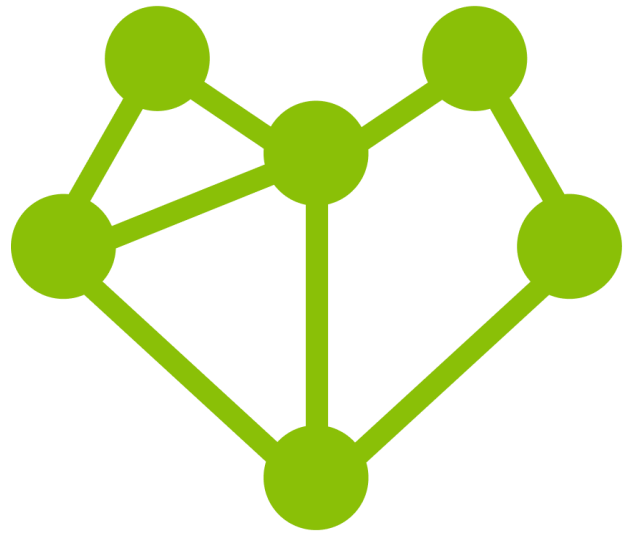
Let's talk about...



Girls'
Programming
Network

I have all the dirt

Let's talk about...



Girls'
Programming
Network

I have all the dirt...y data

I have all the dirt...y data

17 years of:



Student names



Student schools



Tutor names

As well as... parent's names, email addresses, phone numbers, tutor sign ups, role allocations, student survey responses, etc, etc

What's all this data for?



Providing medical and wellbeing support during events



Tracking how students and tutors move through GPN over the years



Mapping where people come to GPN events from

What's wrong with my data?

Let's start with school names

What's wrong with my data?

Let's start with school names

Cerdon College

What's wrong with my data?

Let's start with school names

cerdon college

Cerdon College

What's wrong with my data?

Let's start with school names

cerdon college

Cerdon College

Cerdon

What's wrong with my data?

Let's start with school names

cerdon college

Cerdon College

Cerdon

Cerdon college
Merrylands

Cerdon College
- Merrylands

Cerdon college
in merrylands

What's wrong with my data?

Let's start with school names

cerdon college

Cerdon College

Cerdon

Cerdon college
Merrylands

Cerdon College
- Merrylands

Cerdon college
in merrylands

cerdon collage

Cerdon
colledge

What's wrong with my data?

It get's harder...

It's initialism time!

What's wrong with my data?

It's initialism time...

Olmc

jrahs

Mfhs

CTHS

GRC Hurstville

What's wrong with my data?

It's initialism time...

Olmc

jrahs

Mfhs

CTHS

GRC Hurstville

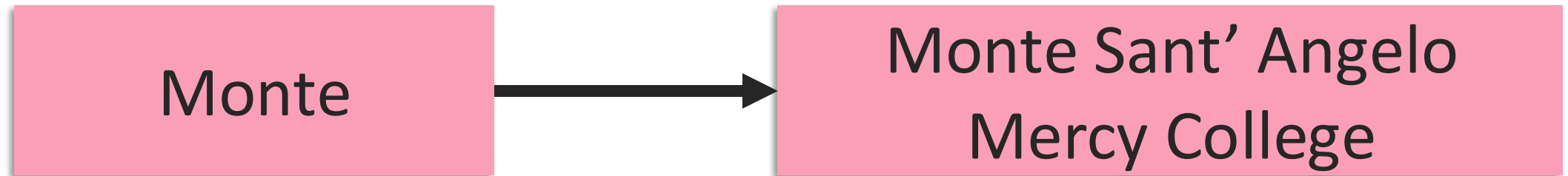
PLC

PLC Sydney

MLC

What's wrong with my data?

And don't get me started on...



Sydney school names
Fake your way with Gen AI

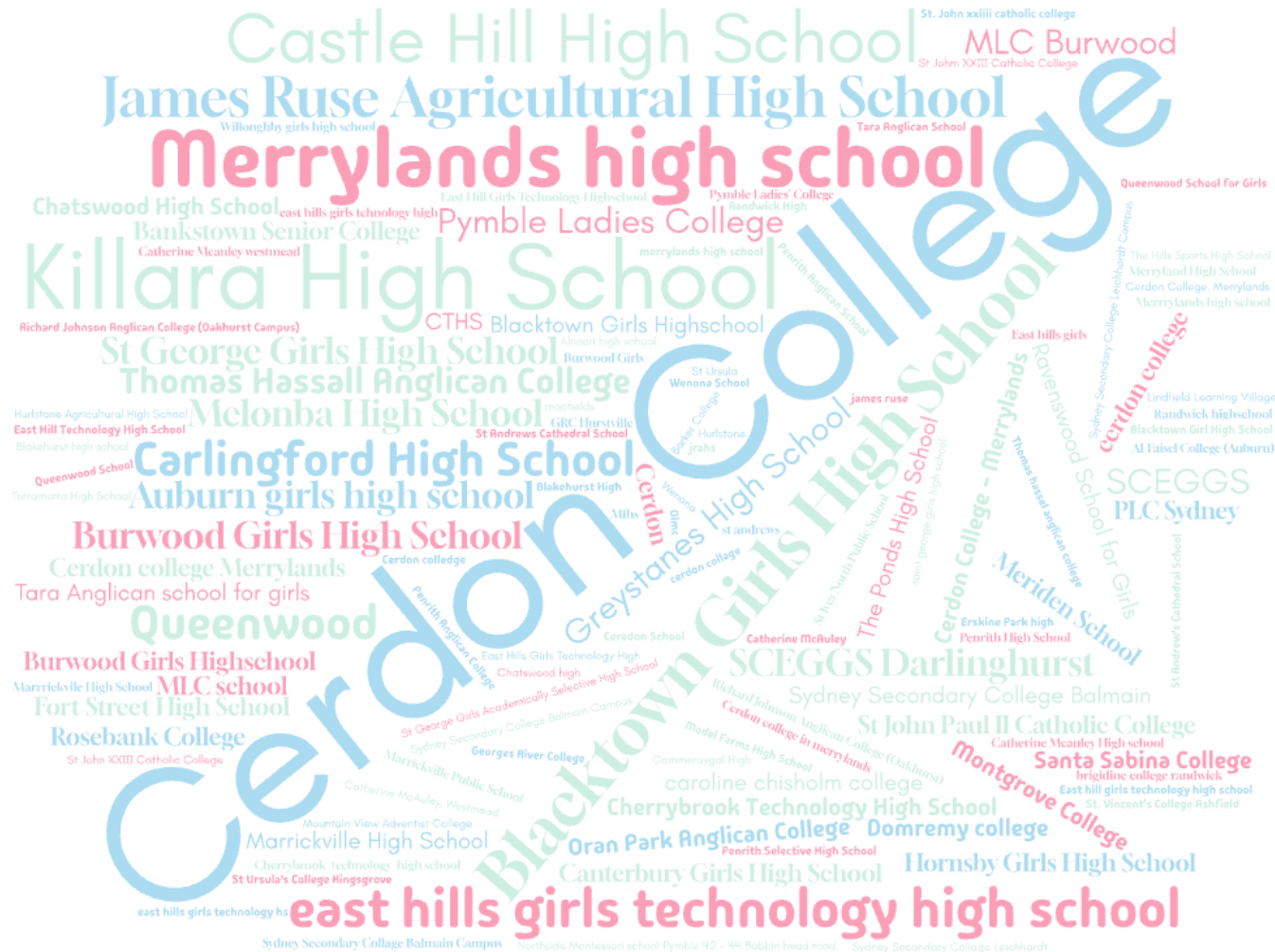
Mapping GPN schools?



Thanks ChatGPT

Correct School Names (Sydney & surrounds)

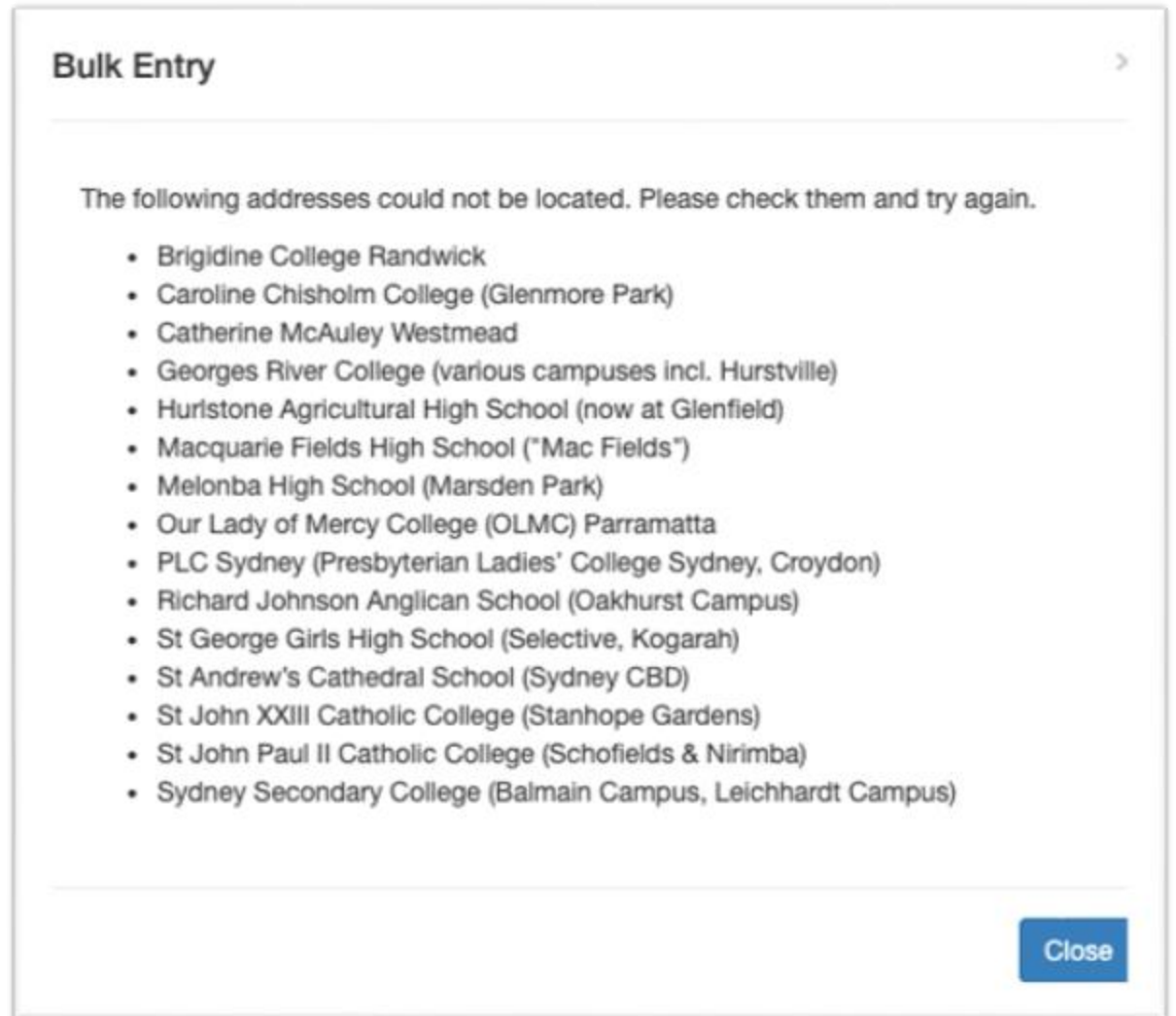
- Al-Faisal College (Auburn)
- Al Noori Muslim School (Greenacre)
- Auburn Girls High School
- Bankstown Senior College
- Barker College (Hornsby)
- Blacktown Girls High School
- Blakehurst High School
- Brigidine College Randwick
- Burwood Girls High School
- Cammeraygal High School
- Canterbury Girls High School
- Carlingford High School
- Caroline Chisholm College (Glenmore Park)



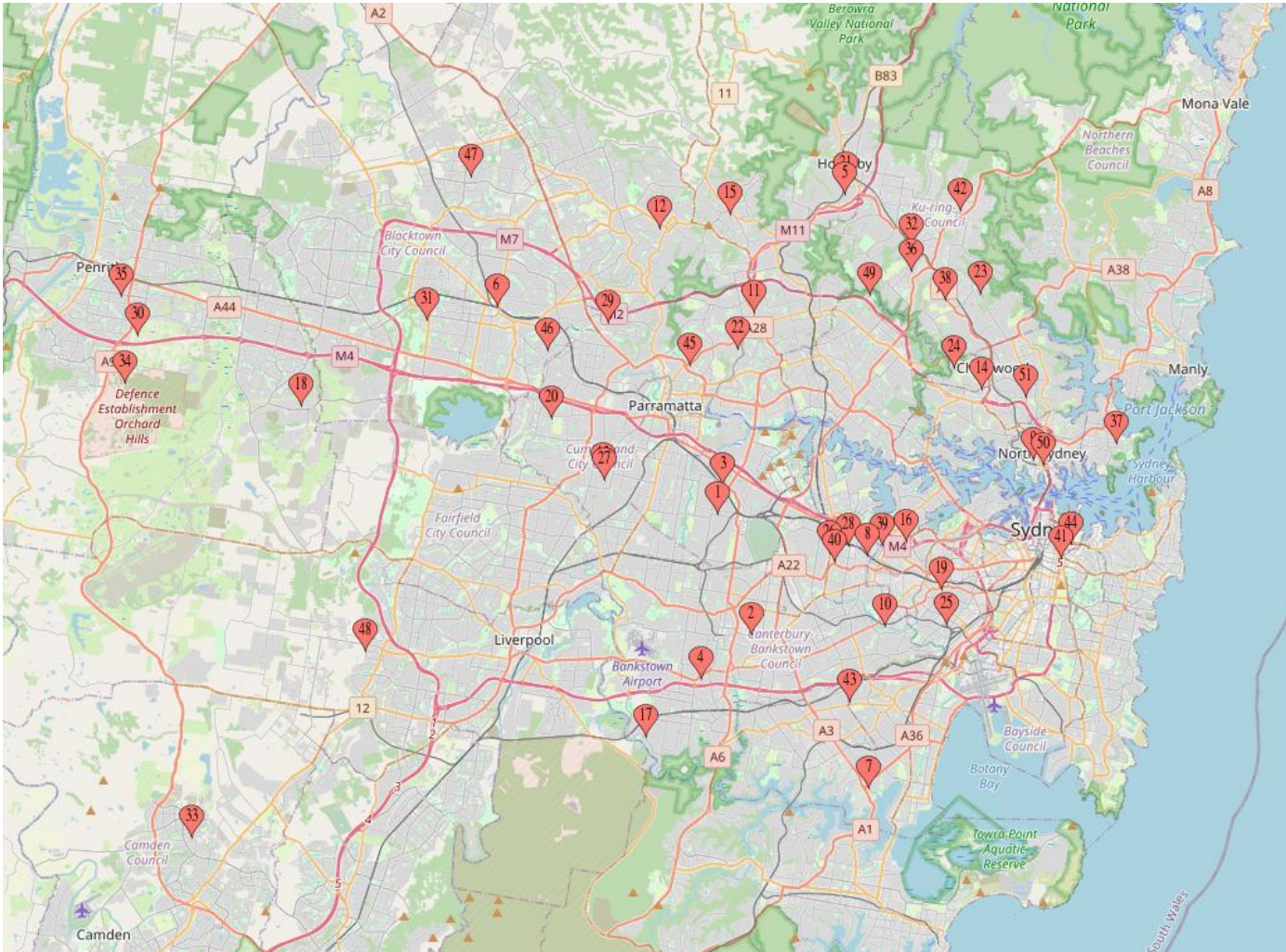
Mapping GPN schools?

It wasn't perfect...

*(maybe that's the
mapping tools fault)*



Mapping GPN schools?



**But it was
good enough!**

Act like a local...in 42 lines of code

Enter a school name: PLC Syd
Presbyterian Ladies' College, Sydney

Enter a school name: PLC north
Pymble Ladies' College

Enter a school name: SACS
St Andrew's Cathedral School

Enter a school name: Kambie
Kambala

```
33 # Initialize Azure OpenAI client with key-based authentication
34 client = AzureOpenAI(
35     azure_endpoint=endpoint,
36     api_key=subscription_key,
37     api_version="2025-01-01-preview",
38 )
39
40 while True:
41     school_guess = input("Enter a school name: ")
42     response = client.chat.completions.create(
43         messages=[
44             {
45                 "role": "system",
46                 "content": "Your task is to correct school names provided by a users to the closest match from a predefined list of schools. Some children use nicknames or abbreviations for the schools and frequently make typos and spelling mistakes, so you should try to match the input to a school they mean based on common usage. The school should be within Sydney, preferably a high school, in NSW, Australia. If it is a nickname it is probably a private school. Only with the exact name of the school, or 'I'm not sure' if you cannot find a match, include your best guess.",
47             },
48             {
49                 "role": "user",
50                 "content": school_guess,
51             }
52         ],
53         max_completion_tokens=800,
54         temperature=1.0,
55         top_p=1.0,
56         frequency_penalty=0.0,
57         presence_penalty=0.0,
58         model=deployment
59     )
60
61     print(response.choices[0].message.content)
```


Comparing fuzzy
matching libraries,
for humans

Why match students?

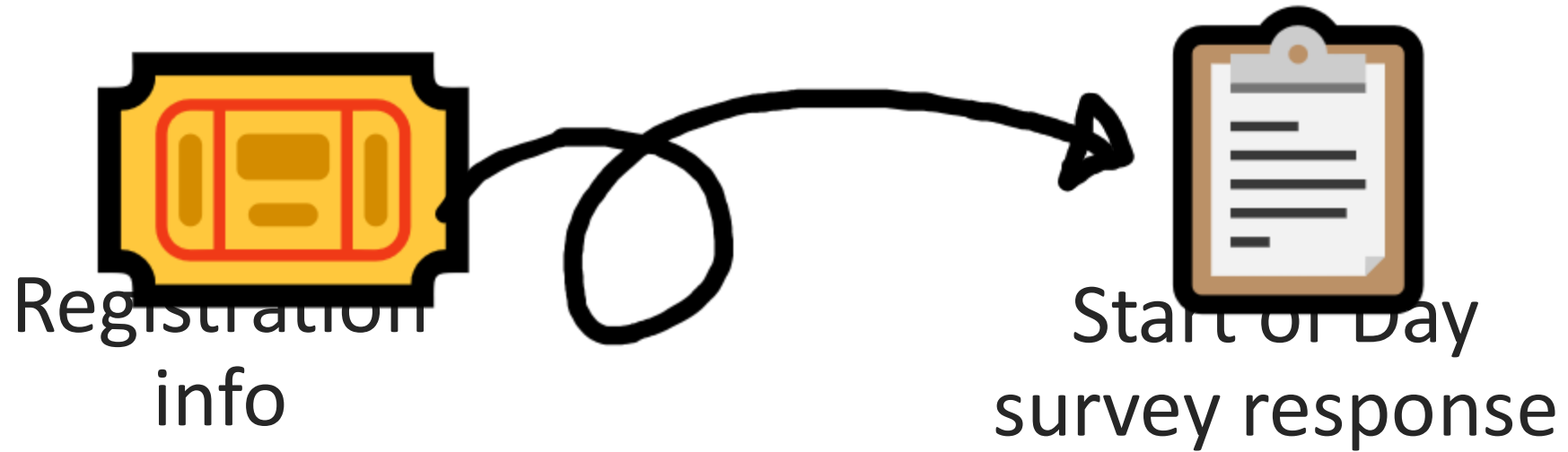


Providing medical and wellbeing support during events



Tracking how students and tutors move through GPN over the years

My Student Data Jigsaw



My Student Data Jigsaw

My Student Data Jigsaw

Elizabeth Wong

What's wrong with my data?

Lizzie (Xiao)
Wong

Elizabeth Lily
Wong

Liz W

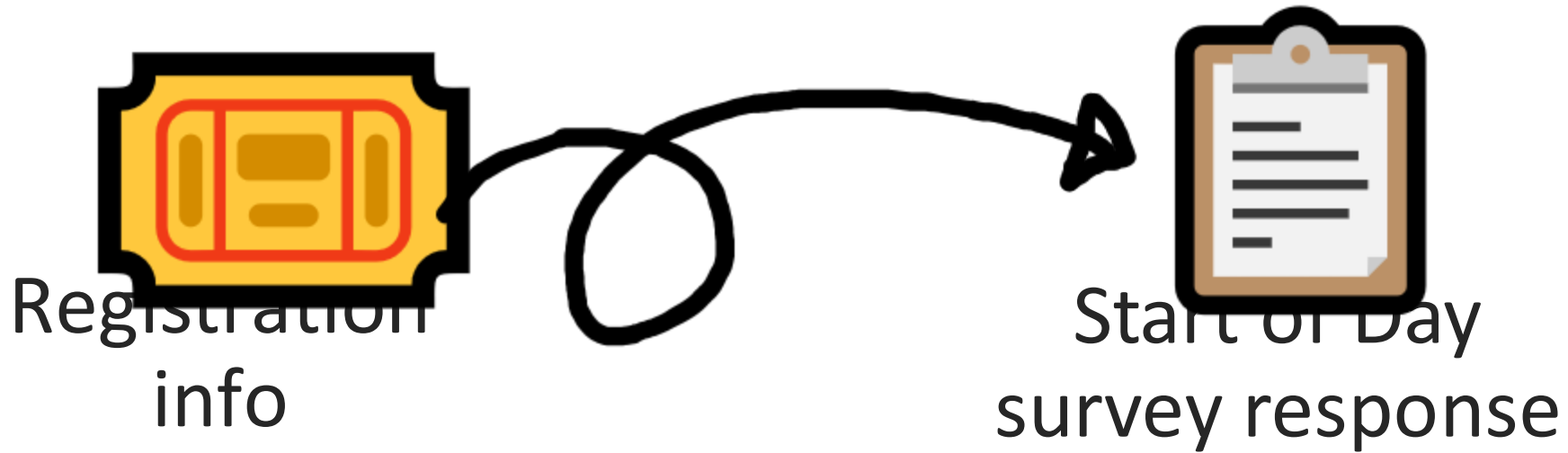
Elizabeth Wong

Lizzie W

Xiao (Liz) Wong

Flame 

My Student Data Jigsaw



Elizabeth Wong
has an EpiPen

Lizzie W *is in the
Pacman classroom*

My Student Data Jigsaw

But unlike school names...

*Kids' full names aren't all
over the internet*

An algorithmic approach

1. Textdistance
2. Python-Levenshtein
3. Nicknames
4. PyNameMatcher
5. Fuzzywuzzy
6. Rapidfuzz

My 100+ Fake Students

- **Generate** correct & mismatched names pairings
- **Different kinds of *variations***
 - Nicknames/informal
 - Case mismatch/typos
 - Partial name
 - Combos of above
- **Australian** multi-cultural selection

Alexandria Young

alex yung

Alexis Young

Alex Youg

sasha young

ALEX YOUNG

Results

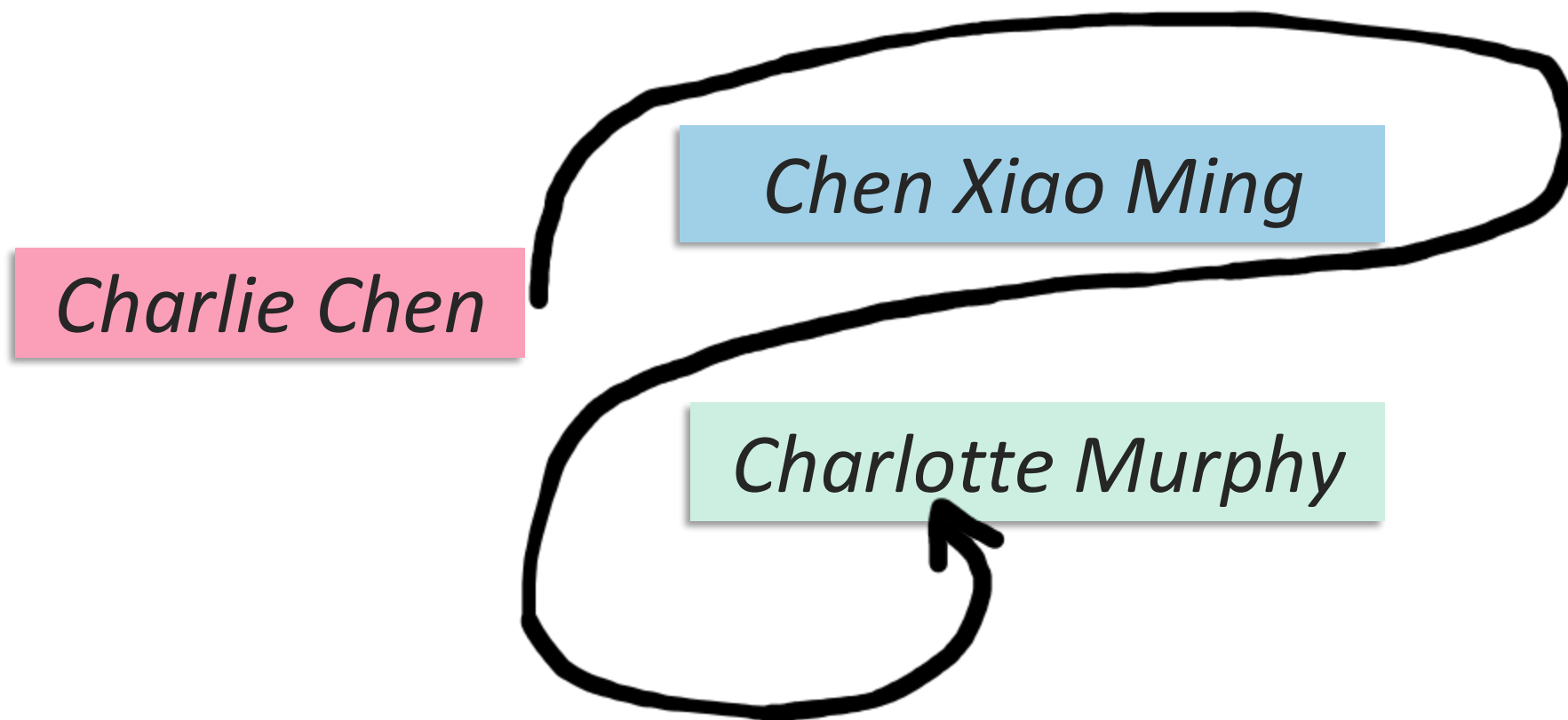
- Total combinations: 317
- Matches by all algorithms: 298 names
(when case corrected)
- Distance-only libraries struggled: 19 names
(when case corrected)
- Nickname-only libraries struggled: 15 names
(prominent from non-anglo names)

An algorithmic approach

Library	Base Accuracy	Case Corrected	Name Fragments
textdistance	93.1	94.0	94.3
PyNameMatcher	93.1	94.0	94.3
python-Levenshtein	93.1	94.0	94.3
nicknames	94.6	95.3	94.6
fuzzywuzzy	99.4	99.4	93.1
rapidfuzz	97.8	99.4	92.4

Results

Names that weren't matched by any algorithm: 1



An algorithmic approach

Library	Base Accuracy	Case Corrected	Name Fragments
textdistance	93.1	94.0	94.3
PyNameMatcher	93.1	94.0	94.3
python-Levenshtein	93.1	94.0	94.3
nicknames	94.6	95.3	94.6
fuzzywuzzy	99.4	99.4	93.1
rapidfuzz	97.8	99.4	92.4

 ***Fastest!***

Rapid Fuzz Results

Names not matched

Charlie Chen

Chen Xiao Ming

Rita Thompson

Margaret Thompson

Sarah Thompson



A hybrid approach for
real data* matching

*All student names changed for slides**

Rapid Fuzz – first pass

Data In:

“Clara Smithe :D”



Student
Submitted
name in survey

[All checked in*
student Full names]



*Repeat for Not checked in students who may have evaded us

Rapid Fuzz – first pass

Data out:

Checked In

“Clara Smithe :D”

“Clara J Smithe”

Score: 95.00

Not Checked In

“Clara Smithe :D”

“Claire Sanders”

Score: 55.45

Rapid Fuzz – first pass

Results (Min match score = 80)

124/128 students
successfully matched
117 exact matches, 7 found matches

No false matches!

Rapid Fuzz – second pass

Some students are signed up without their last name!



“Mia Jessica”

Rapid Fuzz – second pass

Some students are signed up without their last name!



“Mia Jessica”

Buyer first name: “Jason”

Buyer last name: “Williams”

email: jason.williams@gmail.com

School: Chatswood HS

Notes: “Can’t eat Kiwi fruit”

...

Rapid Fuzz – second pass

Some students are signed up without their last name!



Mia Jessica

+

Williams

Maybe we'll get lucky?

Rapid Fuzz – second pass

Results (Min match score = 80)

1 of 4 remaining students
successfully matched

No false matches!

The remaining students

The 3 remaining students couldn't even be human matched



Can Gen AI Help?

Can we use a basic RAG pattern to give GenAI the context to match students?



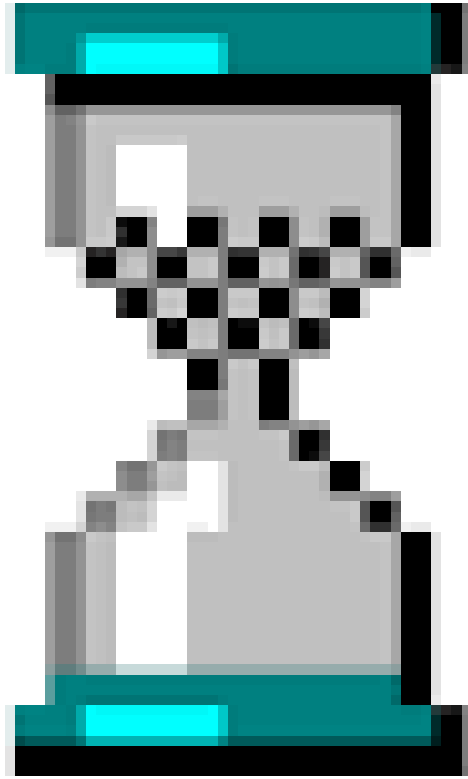
Can Gen AI Help?

```
attendees_info = []  
for i, attendee in enumerate(attendee_list, 1):  
    attendee_name = attendee.get("Student's full name", "")  
    attendee_school = attendee.get("What school does the student attend?", "")  
    attendee_year = attendee.get("What year is the student in at school?", "")  
    attendees_info.append(f"{i}. Name: {attendee_name}, School: {attendee_school}, Year: {attendee_year}")  
attendees_text = "\n".join(attendees_info)
```

Can Gen AI Help?

```
prompt = f"""I need to match a student from a survey with their registration in an attendee list.
.....
.....Survey Student Information:
.....- Name: {survey_name}
.....- School: {survey_school}
.....- Year: {survey_year}
.....
.....Attendee List:
.....{attendees_text}
.....
.....Find the best match for the survey student in the attendee list.
.....Consider name variations, nicknames, and possible misspellings.
.....Return your answer as a JSON object with these fields:
.....- match_index: the number of the matching attendee (1-based index from the list)
.....- confidence: your confidence level (0-100)
.....- reasoning: brief explanation of why this is a match
.....
.....If there's no good match, set match_index to 0 and explain why.
....."""
```

Can Gen AI Help?



Can Gen AI Help?

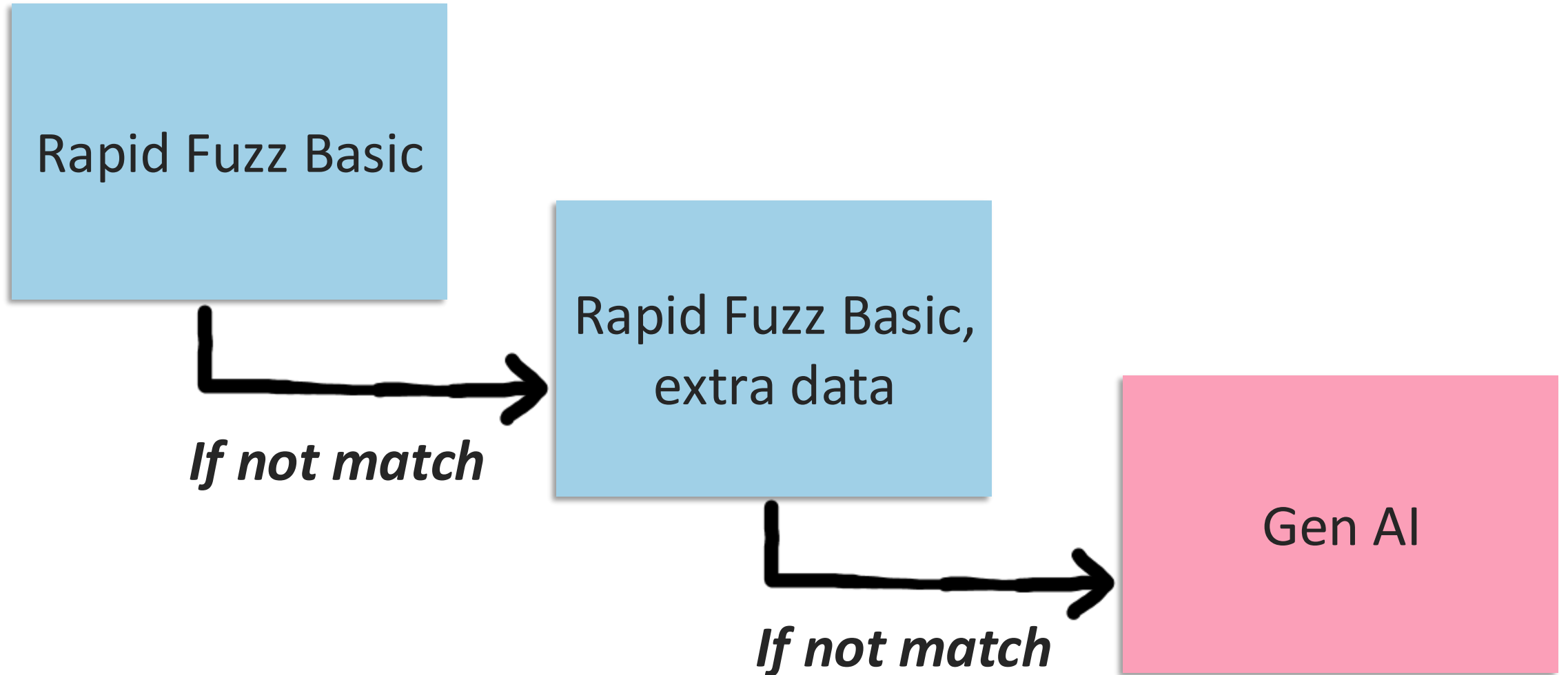
Rapid Fuzz

- 124/128 matched
in *Pass 1*
- 1 Extra matched
in *Pass 2*
- 3 unmatched

GPT5

- 124 matched
(but not the same 124,
finds the Pass 2 name)
- Includes non-
name context
- 4 unmatched

Can Gen AI Help?



What will I
match next?

So far, I've matched students from a single Sydney GPN event

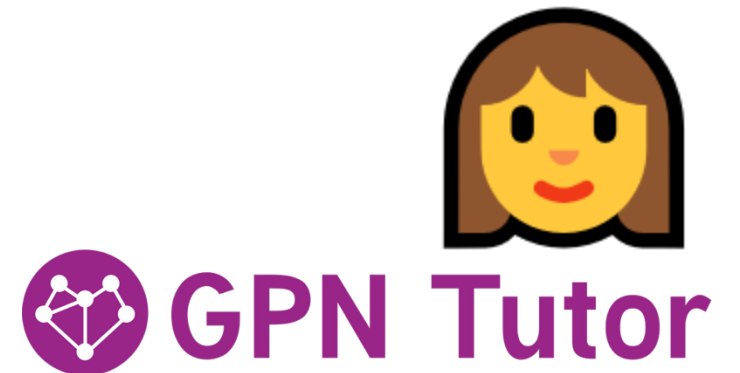
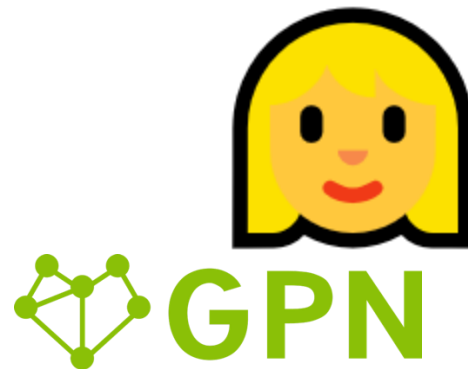
First Up:

- Test more workshop data to find any outliers name styles.
- Test data from workshops around the national GPN network

Second Up

Longitudinal Tracking

- How often does GPN make a difference?
- What is the typical GPN journey?



I'm going to need a bigger data structure...

- 7000+ historic enrolments
- 9 (historic) locations
- A community that moves!

A fast way to match students and tutors GPN journeys around the country is the goal!



Thanks!

Repo: aka.ms/rn-whats-in-a-name

aka.ms/foundry-discord-rn



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